```cpp
// 1. Write a program to implement bubble sort for sorting n elements in an array.

#include <bits/stdc++.h>
using namespace std;

void bubbleSort(int arr[], int n){
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-1-i;j++){
            if(arr[j]>arr[j+1]){
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
}
void printArr(int arr[], int n){
    for(int i=0;i<n;i++) cout << arr[i] << " ";
    cout << "\n";
}

int main(){
    int n;
    cout << "Enter array size: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for(int i=0;i<n;i++) cin >> arr[i];
    bubbleSort(arr,n);
    cout << "Sorted array: ";
    printArr(arr,n);
    return 0;
}
```

```
~/Documents/dslab  master !1 ?13  g++ assignment7_q1.cpp
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
25 10 15 5 30
Enter elements:
Sorted array: 5 10 15 25 30
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 4
2 4 1 3
Enter elements:
Sorted array: 1 2 3 4
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 6
100 50 200 150 25 75
Enter elements:
Sorted array: 25 50 75 100 150 200
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 7
9 7 5 3 1 2 4
Enter elements:
Sorted array: 1 2 3 4 5 7 9
~/Documents/dslab  master !1 ?13
```

```cpp
// 2. Write a program to implement Selection sort for sorting n elements in an array.

#include <bits/stdc++.h>
using namespace std;

void selectionSort(int arr[], int n){
    for(int i=0;i<n-1;i++){
        int mi=i;
        for(int j=i+1;j<n;j++){
            if(arr[j]<arr[mi]) mi=j;
        }
        swap(arr[i],arr[mi]);
    }
}

void printArr(int arr[], int n){
    for(int i=0;i<n;i++) cout << arr[i] << " ";
    cout << "\n";
}

int main(){
    int n;
    cout << "Enter array size: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for(int i=0;i<n;i++) cin >> arr[i];
    selectionSort(arr,n);
    cout << "Sorted array: ";
    printArr(arr,n);
    return 0;
}
```

```
~/Documents/dslab  master !1 ?13  g++ assignment7_q2.cpp
~/Documents/dslab  master !1 ?13  nvim assignment7_q2.cpp
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
25 10 15 5 30
Enter elements:
Sorted array: 5 10 15 25 30
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
1 2 3 4 5
Enter elements:
Sorted array: 1 2 3 4 5
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 7
44 22 11 33 55 77 99
Enter elements:
Sorted array: 11 22 33 44 55 77 99
~/Documents/dslab  master !1 ?13
```

```cpp
// 3. Write a program to implement Insertion sort for sorting n elements in an array.

#include <bits/stdc++.h>
using namespace std;

void insertionSort(int arr[], int n){
    for(int i=1;i<n;i++){
        int x=arr[i], j=i-1;
        while(j>=0 && arr[j]>x){
            arr[j+1]=arr[j];
            j--;
        }
        arr[j+1]=x;
    }
}

void printArr(int arr[], int n){
    for(int i=0;i<n;i++) cout << arr[i] << " ";
    cout << "\n";
}

int main(){
    int n;
    cout << "Enter array size: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for(int i=0;i<n;i++) cin >> arr[i];
    insertionSort(arr,n);
    cout << "Sorted array: ";
    printArr(arr,n);
    return 0;
}
```

```
Sorted array: 11 22 33 44 55 77 99
~/Documents/dslab  master !1 ?13  nvim assignment7_q3.cpp
~/Documents/dslab  master !1 ?13  g++ assignment7_q3.cpp
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 6
170 45 75 90 802 24
Enter elements:
Sorted array: 24 45 75 90 170 802
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
400 500 100 200 300
Enter elements:
Sorted array: 100 200 300 400 500
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 8
121 432 564 23 1 45 788 32
Enter elements:
Sorted array: 1 23 32 45 121 432 564 788
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
400 500 100 200 300
Enter elements:
Sorted array: 100 200 300 400 500
~/Documents/dslab  master !1 ?13
```

```cpp
// 4. Write a program to implement Shell sort for sorting n elements in an array.

#include <bits/stdc++.h>
using namespace std;

void shellSort(int arr[], int n){
    for(int gap=n/2;gap>0;gap/=2){
        for(int i=gap;i<n;i++){
            int temp=arr[i], j;
            for(j=i;j>=gap && arr[j-gap]>temp;j-=gap){
                arr[j]=arr[j-gap];
            }
            arr[j]=temp;
        }
    }
}
void printArr(int arr[], int n){
    for(int i=0;i<n;i++) cout << arr[i] << " ";
    cout << "\n";
}

int main(){
    int n;
    cout << "Enter array size: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for(int i=0;i<n;i++) cin >> arr[i];
    shellSort(arr,n);
    cout << "Sorted array: ";
    printArr(arr,n);
    return 0;
}
```

```
~/Documents/dslab  master !1 ?13  g++ assignment7_q4.cpp
~/Documents/dslab  master !1 ?13  nvim assignment7_q4.cpp
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
1 2 3 4 5
Enter elements:
Sorted array: 1 2 3 4 5
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 8
8 4 2 9 1 5 3 6
Enter elements:
Sorted array: 1 2 3 4 5 6 8 9
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 7
44 22 11 33 55 77 99
Enter elements:
Sorted array: 11 22 33 44 55 77 99
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 7
2 4 10 3 6 9 8
Enter elements:
Sorted array: 2 3 4 6 8 9 10
~/Documents/dslab  master !1 ?13
```

```cpp
// 5. Write a program to implement Radix sort for sorting n elements in an array.

#include <bits/stdc++.h>
using namespace std;

int getMax(int arr[], int n){
    int m=arr[0];
    for(int i=1;i<n;i++) if(arr[i]>m) m=arr[i];
    return m;
}
void countingSort(int arr[], int n, int exp){
    int out[n];
    int count[10]={0};
    for(int i=0;i<n;i++) count[(arr[i]/exp)%10]++;
    for(int i=1;i<10;i++) count[i]+=count[i-1];
    for(int i=n-1;i>=0;i--){
        out[count[(arr[i]/exp)%10]-1]=arr[i];
        count[(arr[i]/exp)%10]--;
    }
    for(int i=0;i<n;i++) arr[i]=out[i];
}
void radixSort(int arr[], int n){
    int mx = getMax(arr,n);
    for(int exp=1;mx/exp>0;exp*=10) countingSort(arr,n,exp);
}
void printArr(int arr[], int n){
    for(int i=0;i<n;i++) cout << arr[i] << " ";
    cout << "\n";
}
int main(){
    int n;
    cout << "Enter array size: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements:\n";
    for(int i=0;i<n;i++) cin >> arr[i];
    radixSort(arr,n);
    cout << "Sorted array: ";
    printArr(arr,n);
    return 0;
}
```

```
Sorted array: 2 3 4 6 8 9 10
~/Documents/dslab  master !1 ?13  nvim assignment7_q5.cpp
~/Documents/dslab  master !1 ?13  g++ assignment7_q5.cpp
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 6
170 45 75 90 802 24
Enter elements:
Sorted array: 24 45 75 90 170 802
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 7
2 4 10 3 6 9 8
Enter elements:
Sorted array: 2 3 4 6 8 9 10
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 5
400 500 100 200 300
Enter elements:
Sorted array: 100 200 300 400 500
~/Documents/dslab  master !1 ?13  ./a.out
Enter array size: 8
121 432 564 23 1 45 788 32
Enter elements:
Sorted array: 1 23 32 45 121 432 564 788
```